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100/100

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Nano-Materials for Sustainable Energy (ChE 494)

Quiz 2: Processing and Characterization of Materials

April 2, 2020 **45 Minutes** **Total: 100**

1. Which of the following is a thin-film deposition technique?

   (A) PVD (B) CVD (c) Electrodeposition (D) All

2. What is the difference between viscous and molecular flow?

   - **Viscous Flow has a relatively higher number of atoms. The mean free path of the molecules is relatively larger than of the object under test. Momentum transfer occurs.**
   - **Molecular Flow has comparatively lesser number of atoms. The mean free path of the molecules is smaller. There is no momentum transfer.**

3. Physical vapor deposition (PVD) can be two types: evaporation and sputtering. What is the difference between thermal and electron-beam evaporation?

   - **In thermal evaporation, the solid material is heated in a closed chamber with high vacuum. Vapour pressure causes the vapour cloud to rise though the chamber and is deposited on the substrate situated higher in the chamber. One major drawback associated with this process is the risk of contamination from other metals.**
   - **Electron Beam Evaporation employs an electron beam, obtained from a cathode filament bombarded on the target solid material, the team is guided by a magnetic field perpendicular to the electron flow, to evaporate it and deposit on the substrate. This too, takes place under vacuum conditions. In contrast to thermal evaporation, the target material is not directly heated in a crucible. This prevents the probability of contamination. It can produce coatings of variable thickness and provides a reduced possibility of heat damage to the substrate.**

4. Write down the steps that must occur in every chemical vapor deposition (CVD) reaction?

   - **Transport of reacting gaseous species to the surface**
   - **Surface Adsorption or Chemisorption of the species on the surface**
   - **Heterogeneous surface reaction catalysed by the surface**
   - **Desorption of the gaseous reaction products**
   - **Transport of reaction products away from the surface**

5. Which of the following transition metals has lowest carbon affinity and why?

   (A) Fe (B) Ni (C) Cu (D) Co

   Copper ([Ar] $3d^{10}$ $4s^{1}$) has the lowest carbon affinity due to its completely filled d-orbitals. The other mentioned transitions metals, all of them have partially filled d-orbitals, with Fe, Co and Ni having 6,7 and 8 electrons in the d-orbitals respectively corresponding to higher carbon affinities.

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6. What is the difference between the CVD growth of graphene on copper (Cu) and nickel (NI)?

- **Copper: The graphene growth on copper follows the surface reaction process. Copper has completely filled d-orbitals, having a low carbon affinity as a result of which the graphene formation takes place on the surface itself. As soon as a layer of graphene if formed, the surface gets fully covered and is no longer exposed to hydrocarbon, thus preventing any further formation of graphene.**
- **Nickel: The graphene growth on copper follows the segregation and precipitation process. Carbon atoms from the hydrocarbon source diffuse into the nickel film, forming a solid NiC solid solution. During cooling, the carbons atoms diffuse out of the solution to the surface, nucleating and forming graphene. Multilayer graphene is usually obtained.**

7. Transition metals and carbon affinity (Arrange with 'decreasing' carbon affinity).

(A) Fe-Co-Ni-Cu (B) Cu-Fe-Ni-Co (C) Cu-Ni-Co-Fe (D) Ni-Cu-Fe-Co

8. During the cooling down process, carbon atoms diffuse out from the NiC solid solution and precipitate on the Ni surface to form graphene films.

(A) True (B) False

9. The graphene growth on Cu is a surface reaction process, which is not self-limiting.

(A) True (B) False

10. Transition metals have partially-filled......................that enable them to adsorb and interact with the hydrocarbons that are used as the carbon sources for the synthesis of graphene.

(A) s-orbitals (B) p-orbitals (C) d-orbitals

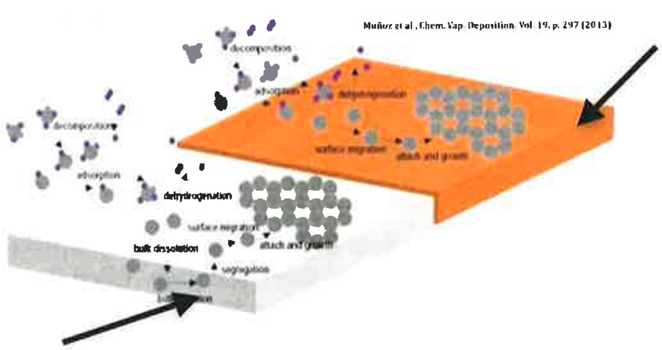
11. Chemical vapor deposition (CVD) is a process where one or more gaseous species react on a solid surface and one of the reaction products is a............................phase material.

(A) Gas (B) Liquid (C) Solid

12. In case of graphene synthesis on metal catalyst substrates *via* CVD, graphene crystallites will seed at positions of locally 'lower chemical potential' such as: ....

(A) Grain Boundary (B) Dislocation and Defects (C) Contaminants (D) All

13. From the following figure: Differentiate between high and low carbon solubility metal catalyst.



Low Carbon Solubility Metal

High Carbon Solubility Metal

14. X-ray photoelectron spectroscopy (XPS) is a surface-analysis spectroscopic technique, which measures the…………………………………………….

    (B) Topography (B) Morphology (C) Elemental Composition (D) All

15. Raman spectroscopy bands arise from a change in the dipole moment (μ) of the molecule due to an interaction of light with the molecule.

    (A) True (B) False

16. When the laser excitation frequency is close with an electronic absorption band of the sample: the intensity enhancement of the Raman scattered light can occur. What is this scattering called?

    (A) Surface-Enhanced Raman Scattering (B) Resonance Raman Scattering (C) All

17. Raman spectroscopy is a spectroscopic technique used to study vibrational, rotational, and other low-frequency modes in a system. It relies on……………………………………scattering (Raman Scattering) of monochromatic light, usually from a laser in the visible, near infrared or near ultraviolet range.

    (A) Elastic (B) Inelastic (C) Both Elastic and Inelastic

18. Scanning electron microscopy (SEM) can be employed for the following sample information.

    (A) Topography (B) Morphology (C) Compositional Information (D) All

19. What is the typical resolution of electron microscopes (SEM & TEM)?

    (A) 200 nm (B) 200 µm (C) 0.2 nm

20. Raman techniques are useful for graphene because the absence of a ………………………………makes all wavelengths of incident radiation resonant, thus the Raman spectrum contains information about both atomic structure and electronic properties.

    (A) sp^3 hybridizations (B) Bandgap (C) Defects (D) All

21. Scanning tunneling microscopy (STM) is applicable when both the sample and tip are conducting.

    (A) True (B) False

22. The resultant dipole moment of a molecule is expressed in the following equation. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering frequencies/components.

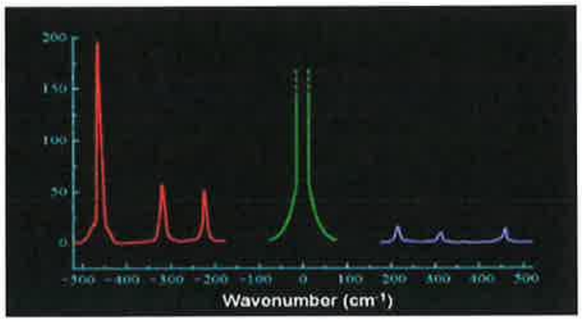
$$\mu = [\alpha_0 E_0 \cos(2\pi \nu t)] + ½ (\delta\alpha/\delta Q)_0 E_0[\cos(2\pi(\nu - \nu_{vib})t) + \cos(2\pi(\nu + \nu_{vib})t)]$$

- **$[\alpha_0 E_0 \cos(2\pi\nu t)]$ - Rayleigh Scattering**
- **$1/2\, (\delta\alpha/\delta Q)_0 E_0[\cos(2\pi(\nu - \nu_{vib})t) + \cos(2\pi(\nu + \nu_{vib})t)]$ - Raman Scattering**
- **$(\nu - \nu_{vib})$ - Stokes Raman Scattering**
- **$(\nu + \nu_{vib})$ - Anti-Stokes Raman Scattering**

23. Scanning electron microscope (SEM) uses a focused……………………………………probe to extract the structural and chemical information of the sample including 2D materials.

    (A) Photon (B) Electron (C) Neutron (D) Phonon

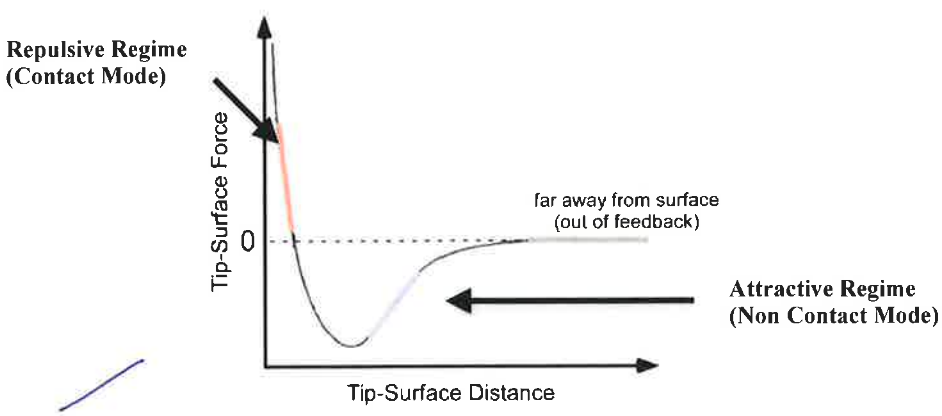
24. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering from the following Raman spectrum. Explain the reason behind the higher intensity for Raman Stokes scattering in contrast to the Raman Anti-Stokes scattering.



Explain: **Red - Raman Scattering Stokes**
Green - Rayleigh Scattering
Blue - Raman Scattering Anti-Stokes

The Stokes scattered light has a low frequency, as the molecule does not reach the original ground state ($\nu-\nu_{vib}$). In contrast, Anti-Stokes scattered light has a relatively higher frequency as the molecule starts at an excited energy level, but reaches the original ground state ($\nu+\nu_{vib}$). Stokes, having a lower energy state, has a higher initial state population. Thus, Raman Stokes Scattering has a higher intensity than Raman Anti-Stokes Scattering.

25. The typical van der Waals force curve for atomic force microscope (AFM) operation mode is presented in the figure below. Show various force-regime and mode of operations?



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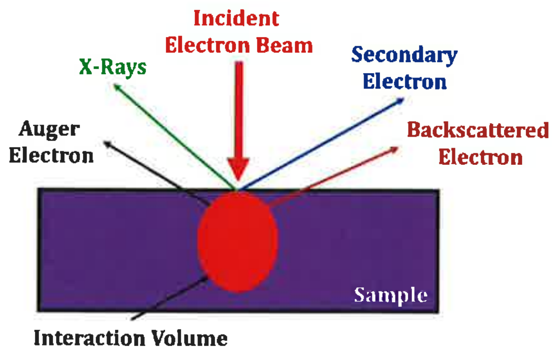
Some Additional Questions:

26. The electron-solid interactions in a scanning electron microscope (SEM) is described in the following figure. Explain the origin of secondary electrons and X-rays?



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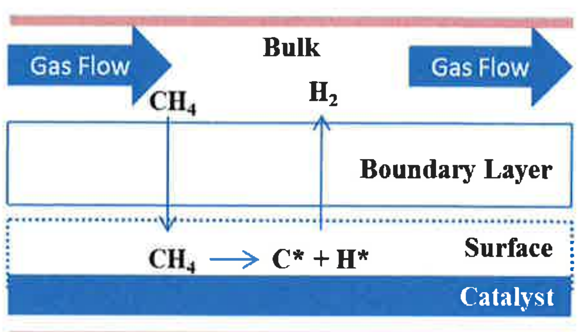
Secondary Electrons originate from the surface or near-surface region of the sample as a result of the interaction between the incident electron beam and the sample. They have a lower energy than the Backscattered Electrons.
X-rays are produced when either the incident electron beam or the backscattered electrons strike the metal surface with sufficient energy to excite X-radiation.

27. Consider the synthesis of high-quality graphene on a catalytic transition metal surface with electronic configuration ([Ar] $3d^{10}4s^1$). As shown in the following figure, there are two fluxes of the active species that coexist: (i) flux of active species through the boundary layer (mass transport region) and (ii) the rate at which the active species are consumed at the surface of the catalyst (surface reaction region) to form the graphene lattice. State the role of kinetic factors on the synthesis of graphene.



Answer:

27) Two fluxes -

- Flux of Active species through the boundary layer (mass transport region)
- Rate at which active species are consumed at the catalyst surface (surface reaction region)

Mathematically,

$F_{MT} = h_g (C_g - C_s) \longrightarrow ①$

$F_{SR} = K_s C_s \longrightarrow ②$

where,

F_{MT} = Flux due to Mass Transport

F_{SR} = Flux due to Surface Reaction

h_g - Mass Transfer Coefficient

C_g - Concentration of Gas in the Bulk

C_s - Concentration of Active Species at the Surface

K_s - Surface Reaction Constant

The slower step is the rate-determining step

At Steady State,

$F_{Total} = F_{MT} = F_{SR} \rightarrow ③$

$\therefore h_g (C_g - C_s) = K_s C_s$

$h_g C_g - h_g C_s = K_s C_s$

$h_g C_g = h_g C_s + K_s C_s$

$h_g C_g = (h_g + K_s) C_s$

$\Rightarrow \frac{h_g C_g}{h_g + K_s} = C_s \longrightarrow ④$

Substituting the value of C_s obtained in ④ in ②

$F_{SR} = K_s C_s = K_s \cdot \frac{h_g C_g}{h_g + K_s} = \frac{K_s h_g}{K_s + h_g} \cdot C_g$

From ③:

$F_{Total} = F_{MT} = F_{SR}$

$\therefore F_{Total} = \frac{K_s h_g}{K_s + h_g} \cdot C_g$

$h_g >> K_s \rightarrow$ Surface Reaction Controlled Region

$h_g \sim K_s \rightarrow$ Mixed Region

$h_g << K_s \rightarrow$ Mass Transport Limited Region